1 The 7 Vs of Big Data

The TLC High-Volume For-Hire Vehicle (HVFHV) records are a seven-year, individually resolved account of app-dispatched urban mobility.

Table 1: The 7 Vs applied to the NYC TLC HVFHV Trip Record Data

V	Evidence in our dataset	Why it matters here
Volume	Raw 37.58 GB across 89 monthly Parquet files (Feb 2019–Jun 2026); June 2026 alone holds 20.78 million trips.	A single month is the practical ceiling for in-memory `pandas` work; the $\geq$12 GB working and $\geq$3 GB processing sets exceed it, so Part 2 requires partitioned, distributed (PySpark) execution.
Velocity	$\sim$692,500 trips/day in June 2026 ($\approx$**8 dispatches/s**); each record carries four event timestamps (request, on-scene, pickup, dropoff).	Fast at source but published monthly: the data suits month-partitioned batch ingestion and retrospective event-time analysis, not live dispatch.
Variety	25 variables of mixed kind: timestamps; continuous measures (`trip_miles`, `trip_time`); money fields (`base_passenger_fare`, `driver_pay`, `tips`); Y/N flags; license codes (Uber HV0003, Lyft HV0005); zone IDs (`PULocationID`/`DOLocationID`).	Heterogeneity drives pre-processing cost: the money fields are not comparable (fare excludes tolls/tips/taxes; `driver_pay` is net of commission), flags and codes are categories, and zone IDs are meaningless until joined to the 265-zone lookup and shapefile.
Veracity	TLC disclaims accuracy. June 2026 checks: 4,844 zero-distance, 6,750 negative-fare and 2 non-positive-duration trips; 14,421 records (**0.07%**) excluded, leaving 20,761,447.	The defects are systematic, not noise: negative fares (refunds/voids) flip the sign of fare aggregates, and zero-distance trips corrupt speed statistics, so records are filtered under a documented rule.
Variability	*Schema:* fare/`driver_pay` columns added May 2022; `cbd_congestion_fee` only from Jan 2025. *Behaviour:* mean distance 7.13 mi at 05:00 vs 4.70 mi at 15:00 (**+51.7%**); Saturday +19.3% trips over Wednesday.	Neither schema nor behaviour stays constant: analyses are only valid where the columns exist, and one 89-month average would mix COVID-19, pre- and post-congestion-pricing periods, so results must be computed per period.
Visualization	Borough pickups (Manhattan 7.14m, Brooklyn 5.72m, Queens 4.77m, Bronx 2.79m, Staten Island 0.35m); hourly distance/duration profiles; LaGuardia 421,225 vs JFK 356,638; Uber 73.4% / Lyft 26.6%; 265$\times$265 OD matrix.	No reader can inspect 20.78m rows, so findings are delivered as graphics; the zero-distance and negative-fare defects were first visible as artefacts in the plotted distributions.
Value	`driver_pay`, `tips` and wait times support effective-earnings analysis; request-vs-match flags measure pooling and unmet wheelchair-accessible demand. **65.6% of pickups occur outside Manhattan.**	Demand is strongest where the subway is weakest, informing transit-gap, curb-space and equity decisions; earnings fields quantify driver welfare; and the Jan 2025 congestion-fee column gives a natural before/after policy window.